Affective Computing

...

A gentle introduction to the study of emotions

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WS Tangible Interactions
Domus Academy
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Presentations

Interaction developer

Member

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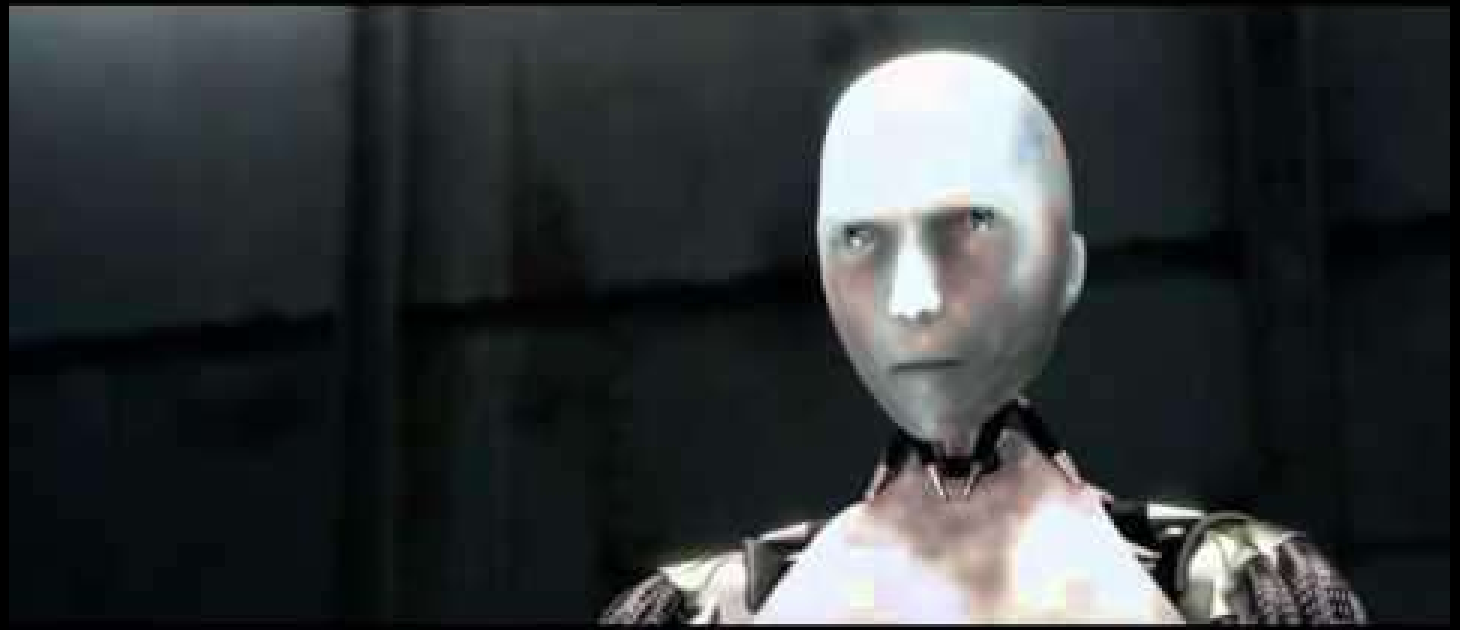
UNIVERSITÀ DEGLI STUDI DI MILANO

PHUSE Lab
Perceptual computing and HUman SEnsing



AIM

Give you some basic knowledge about **Affective Computing** and how this could enhance the effectiveness of a **human-machine interaction**.

AIM

Design, prototype and produce machines that:

- **Detect** emotions
- **Express** emotions
- “**Feel**” emotions

Reference

Picard, R. (1997).
Affective computing. 1st ed.
Cambridge, Mass.: MIT Press.

“The question is not whether intelligent machines can have any emotions, but whether machines **can be intelligent without emotions**”

Marvin Minsky, *The Society of Mind* (1958)

Intro

Are emotions really needed?

Short answer

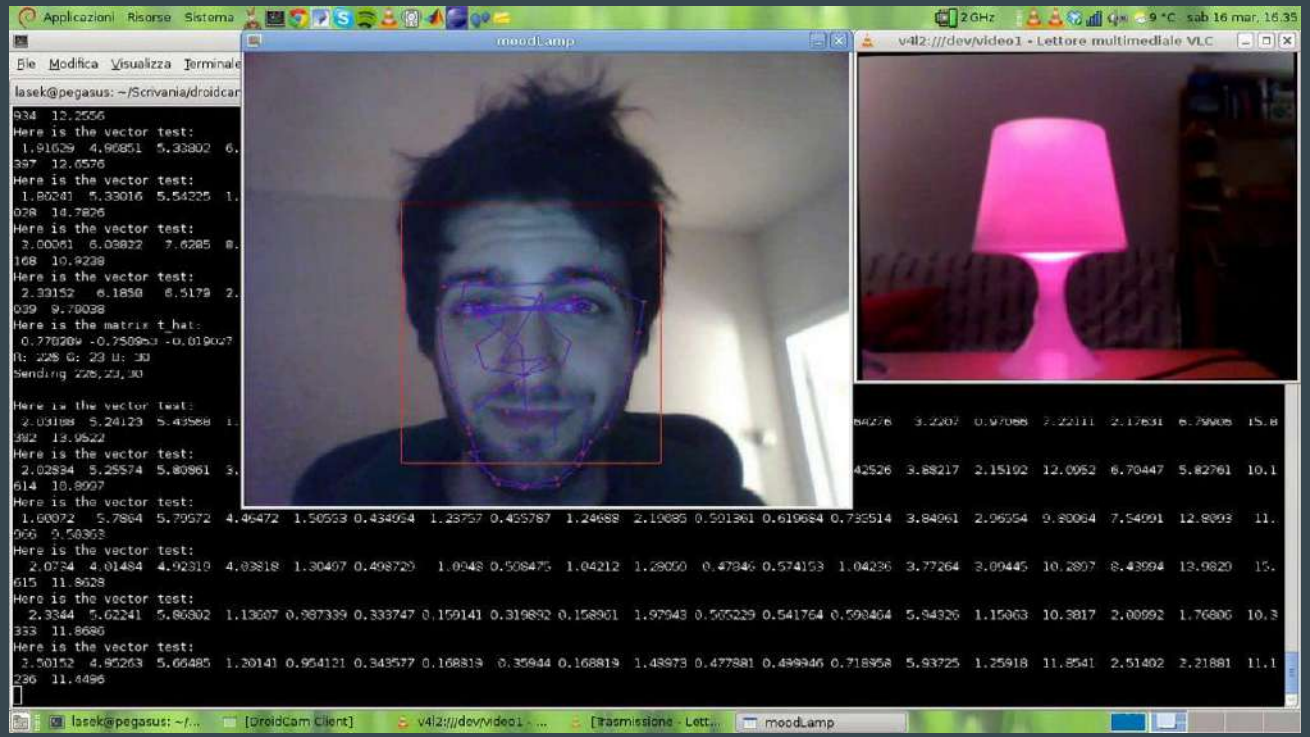
No.

Long answer

Emotions are not a panacea and is not need to be put into everything that computes. Designers should not abuse of it to make computers and other devices affective.

Printers, **lamps** and **moka** works fine without emotions. While others, for example software agents that interact with people, will benefit from a repertoire similar to our own.

AutoEmotive (MIT)



MoodLamp (Università degli Studi di Milano)

NAO (Aldebaran Robotics)

Affective Computing (AC) is an interdisciplinary field spanning computer science, psychology, and cognitive science.

“AC is computing that relates to, arises from, or deliberately influences emotion or other affective phenomena” (Picard, 1997)

The machine should interpret the emotional state of humans and adapt its behaviour to them, giving an appropriate response for those emotions.

Affective computing

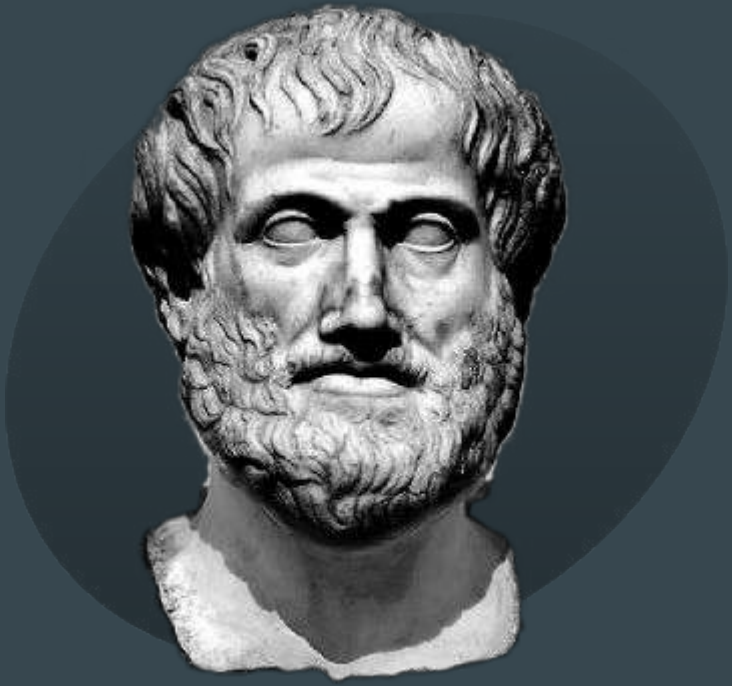
...but, wait!

What is an emotion?

Emotions are

- intentional, representational and part of virtue.
  (Aristotle, 330 A.D.)

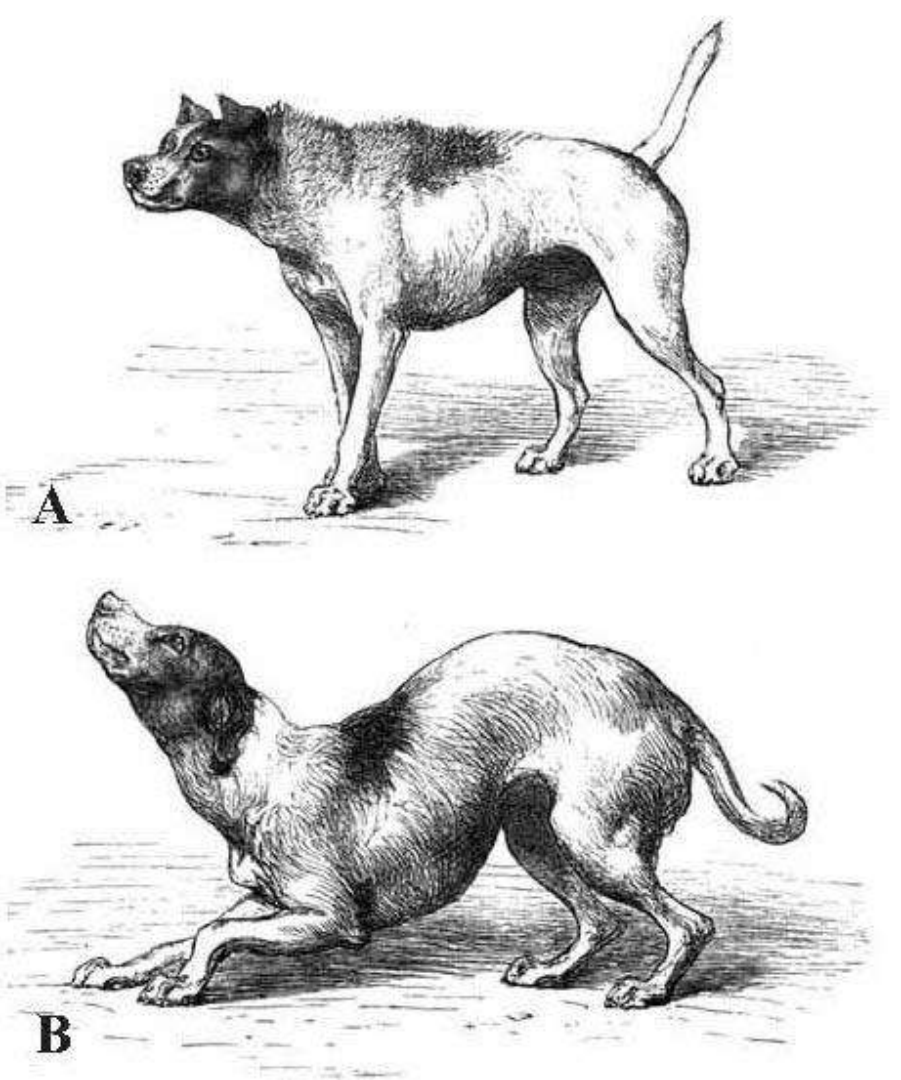
- an obstacle to reason and therefore an obstacle to virtue.
  (Stoicism, 300 A.D.)



Emotions are

- the result of evolution, served in communication and survival.
  (Charles Darwin, 1800)

- physiological response to a stimuli.
  (William James, 1884)



Emotions are

- discrete and expressed by a set of facial expressions.
  (Paul Ekman, Carroll Izard)

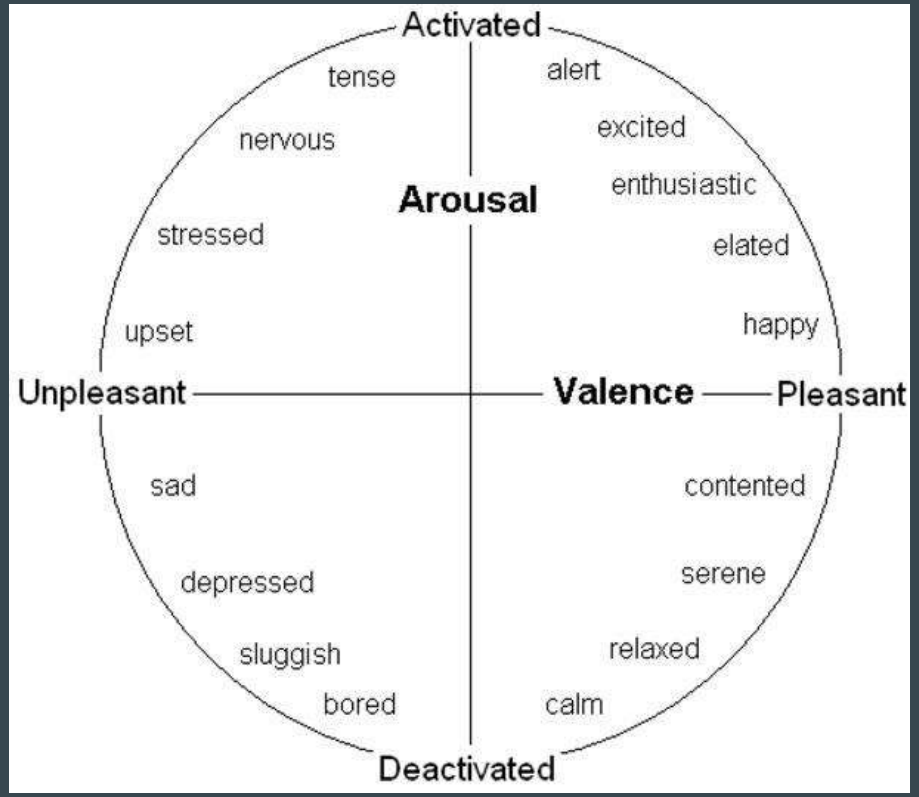
Emotions are discrete

Universality of basic facial expressions.

(Ekman, 1971;1992; 1993)

Emotions are

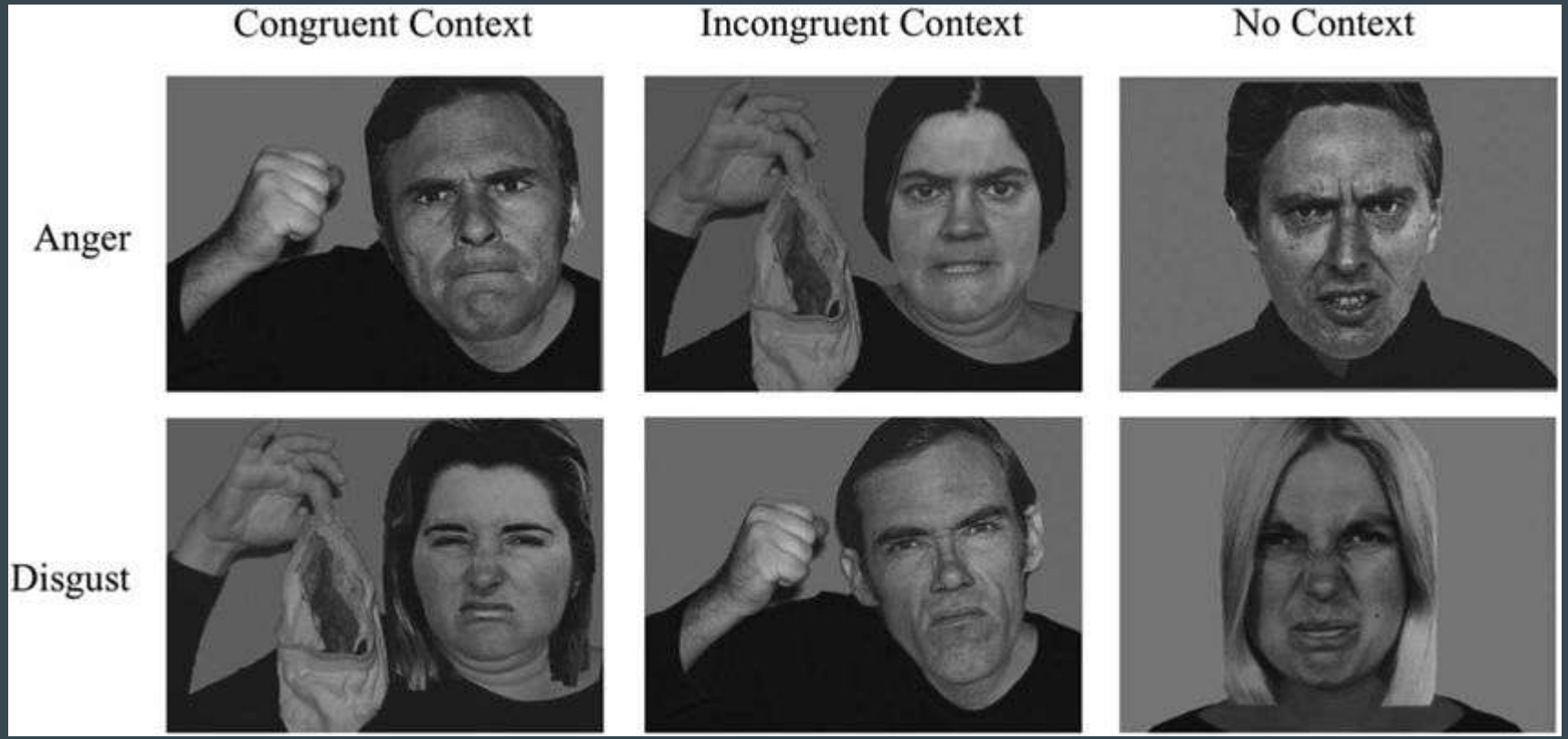
- influenced by a core affect and expressed in terms of valence and arousal.
  (James Russell)



Emotions are dimensional

There is no one-to-one correspondence between an emotion word and a facial expression.

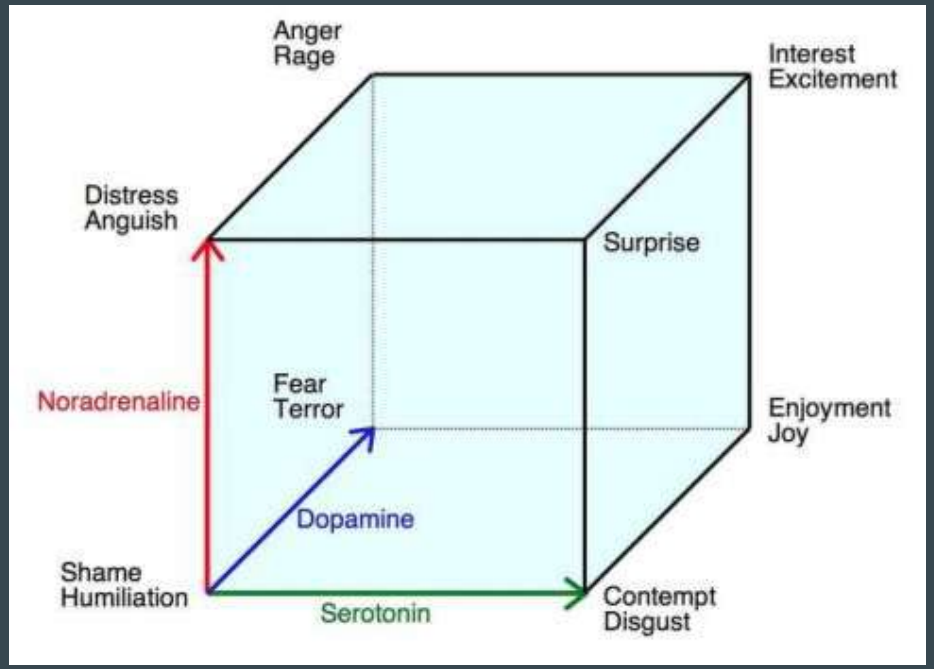
Emotions are dimensional



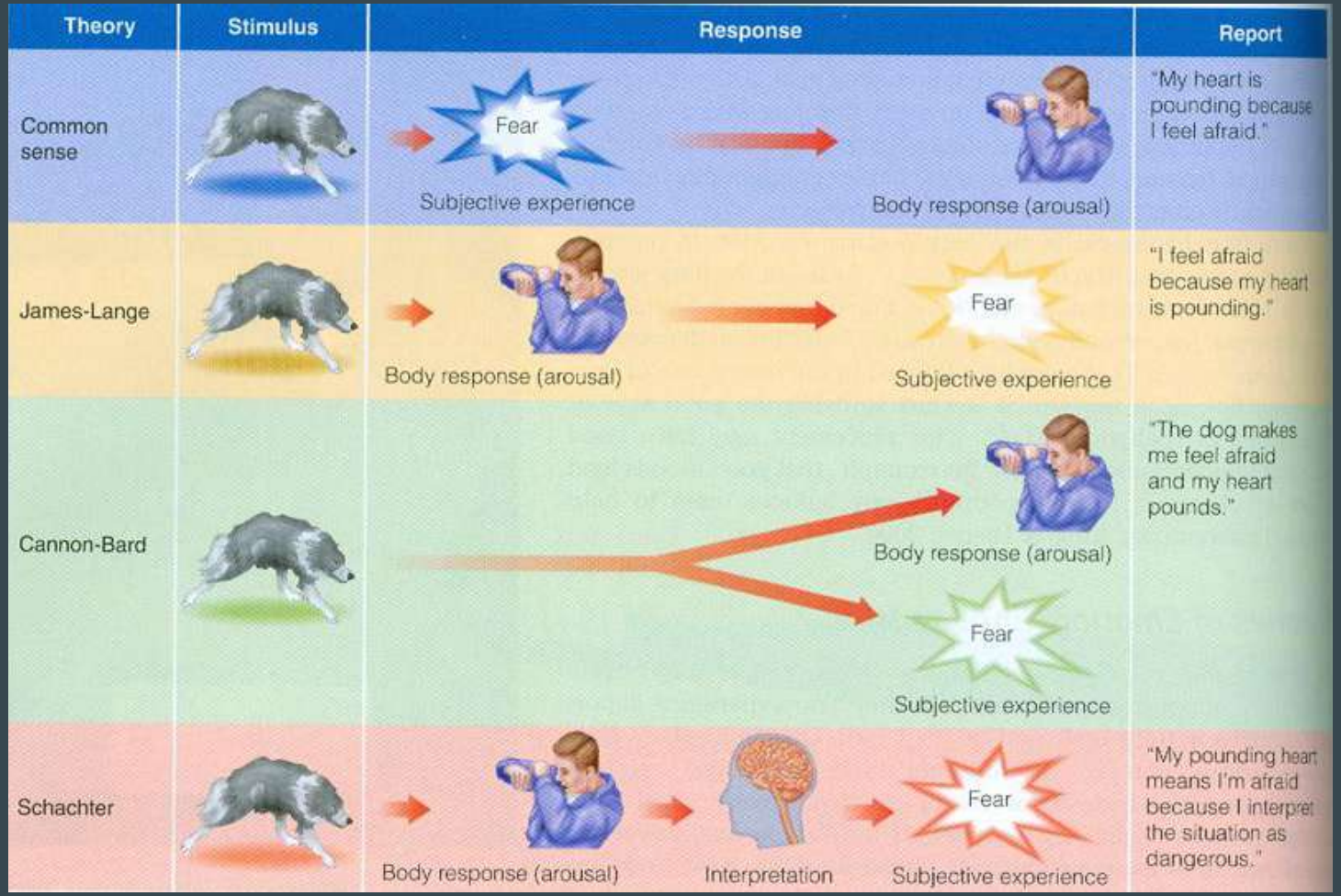
Emotions are dimensional

The emotions are neither discrete entities nor points on a few dimensions; they are overlapping point-clouds in an N-dimensional space.

(Nesse, Ellsworth)



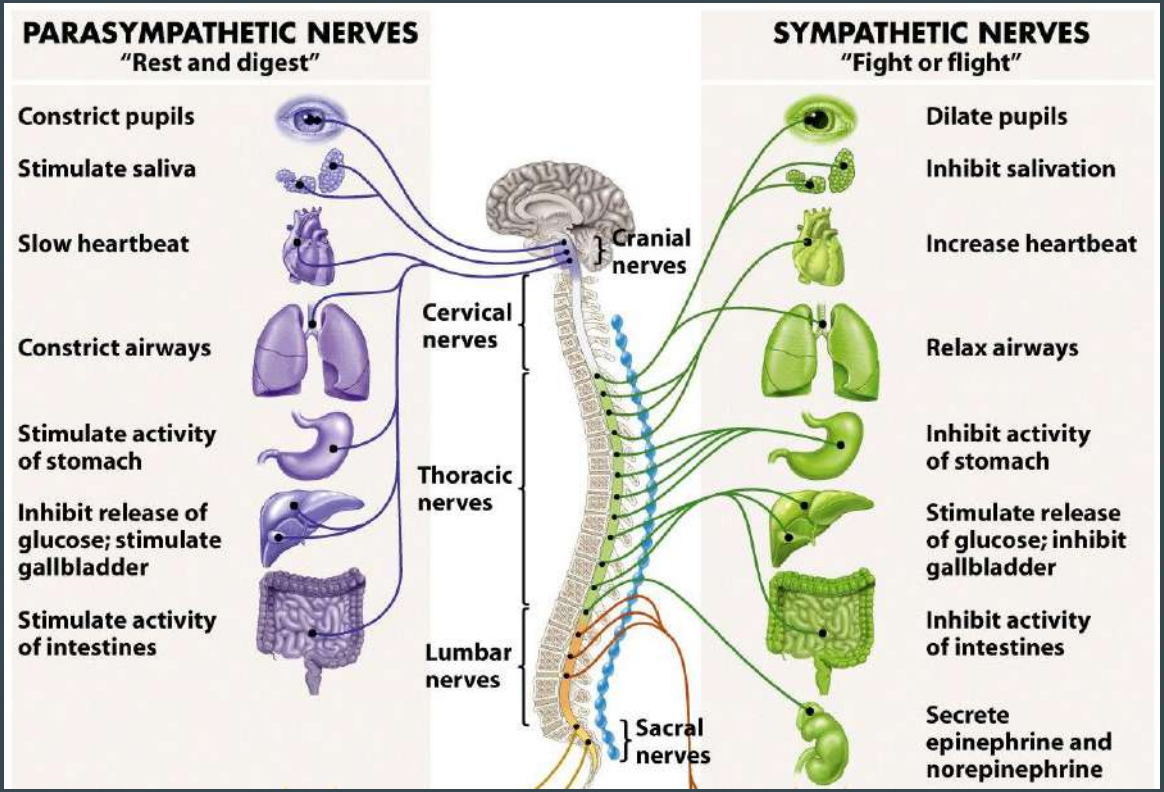
How emotions are expressed?


Theory
Stimulus
Response
Report
Common sense
Fear
Subjective experience
Body response (arousal)
"My heart is pounding because I feel afraid."
James-Lange
Body response (arousal)
Fear
Subjective experience
"I feel afraid because my heart is pounding."
Cannon-Bard
Body response (arousal)
Fear
Subjective experience
"The dog makes me feel afraid and my heart pounds."
Schachter
Body response (arousal)
Interpretation
Fear
Subjective experience
"My pounding heart means I'm afraid because I interpret the situation as dangerous."

Emotional cues

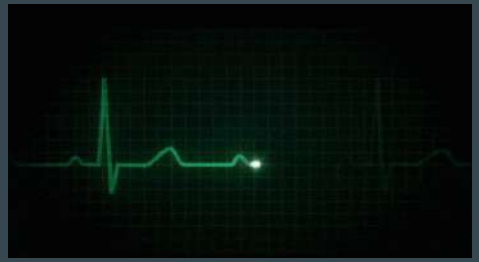
- Visible
  - Facial expression
  - Voice intonation
  - Gesture
  - Posture
  - Pupillary dilation
- Less Visible
  - Respiration
  - Heart rate
  - Temperature
  - Electrodermal response
  - Muscle actions
  - Blood pressure

Affective computing



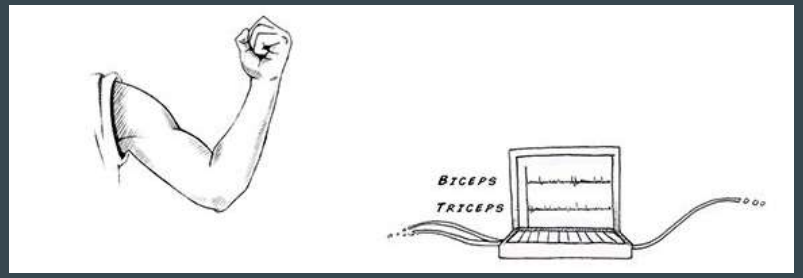
Autonomic Nervous System (ANS)

- Heart rate (HRV)

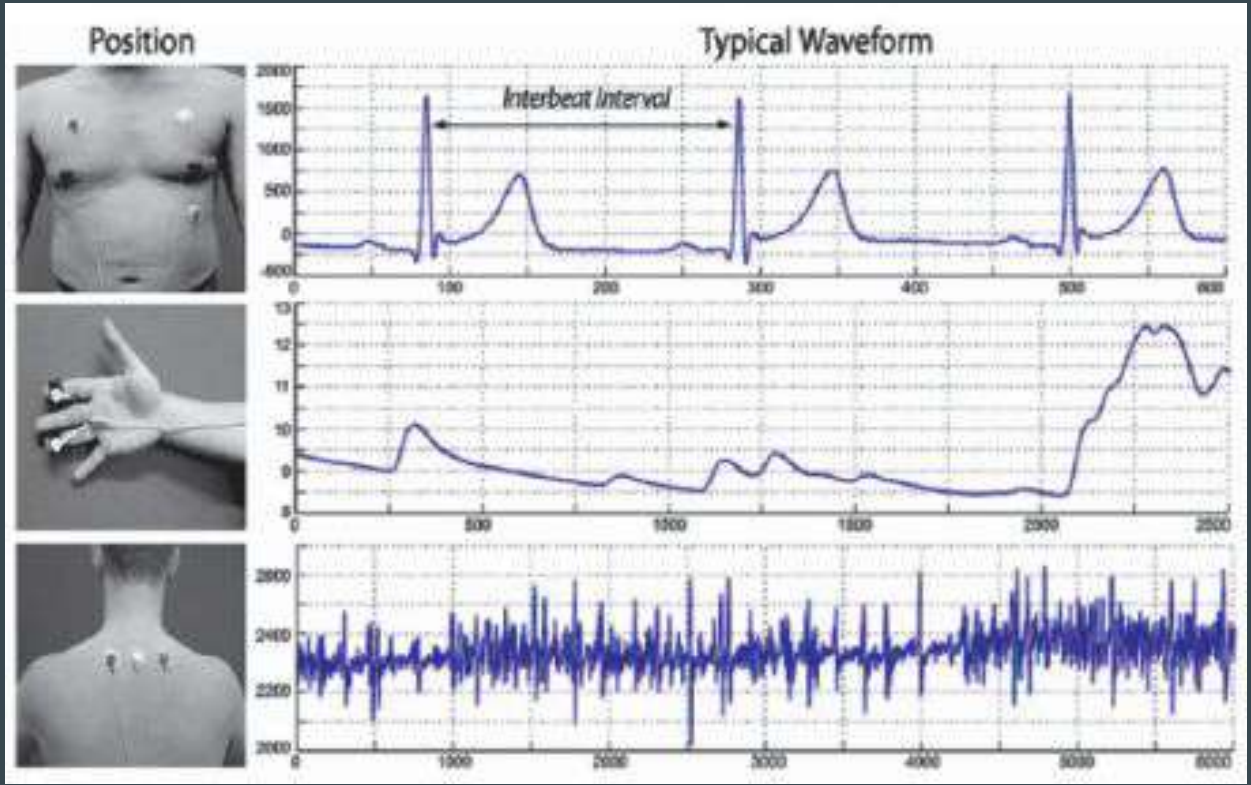


- Electrodermal response (GSR)

- Muscle activity (EMG)



Measure of physiological signals



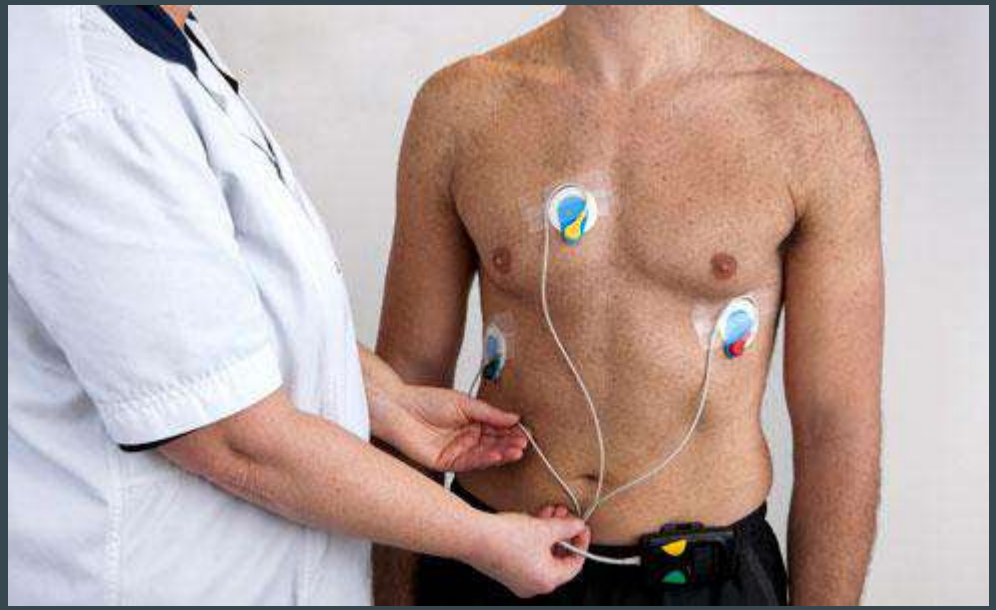
Measure of physiological signals

Heart rate variability (HRV) refers to the oscillation of the interval between consecutive heartbeats

Measure of physiological signals - HRV

HRV is obtained through the Electrocardiography (ECG).

... typically invasive!

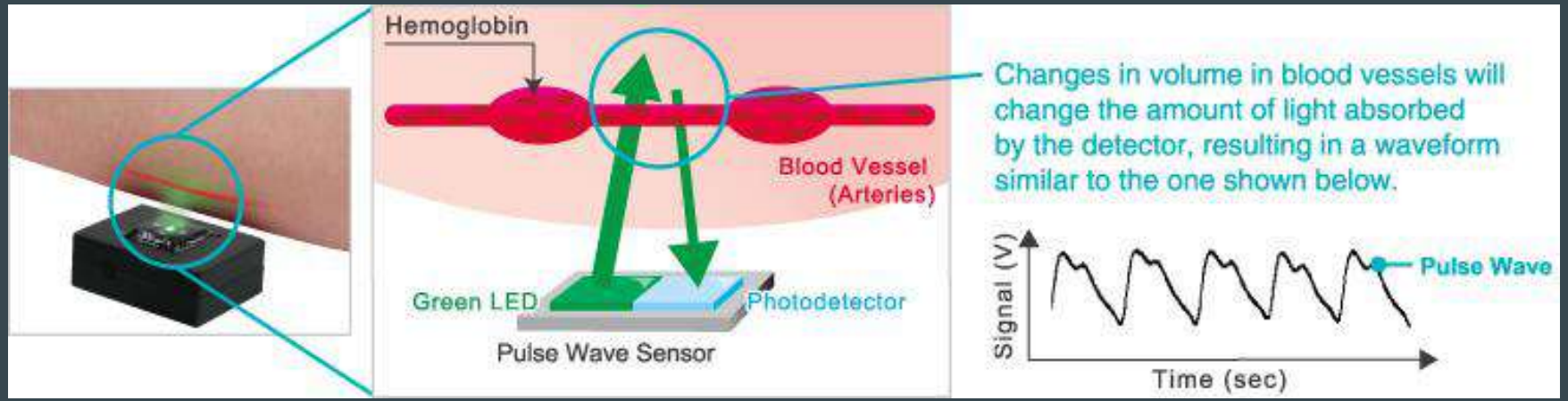


Measure of physiological signals - HRV

Blood volume pulse (BVP) measures indirectly the heart rate and is less invasive.

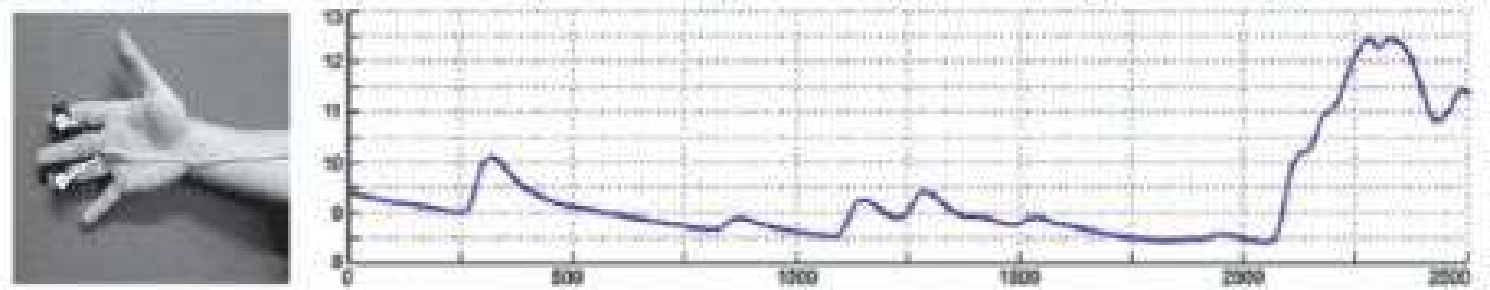
Measure of physiological signals - HRV

Sends infrared light with a specific wavelength (990nm) and measures the reflected amount of light.



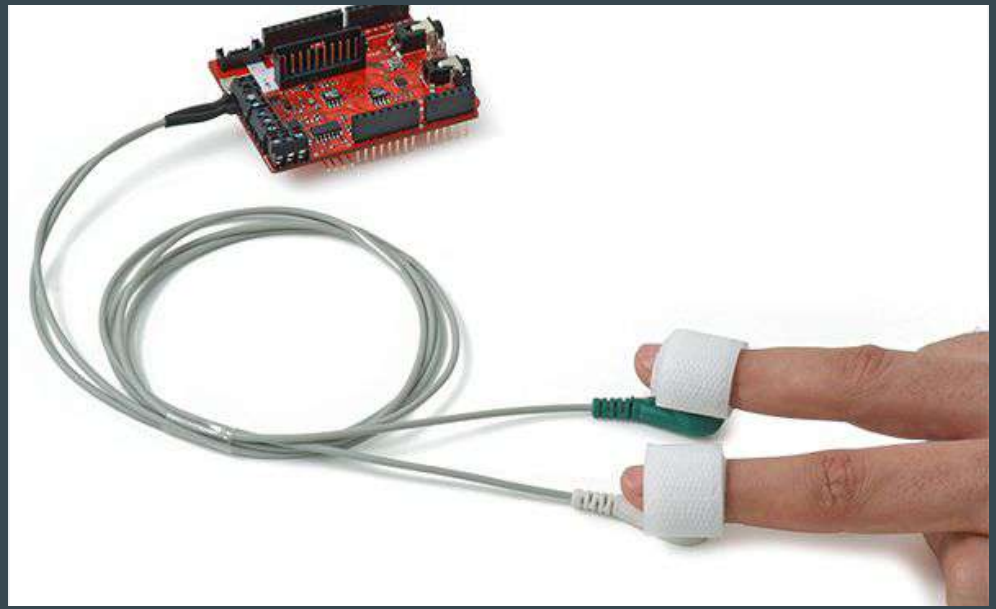
Measure of physiological signals - HRV

Skin conductivity (SC) sensor measures the skin's ability to conduct electricity.



Measure of physiological signals - GSR

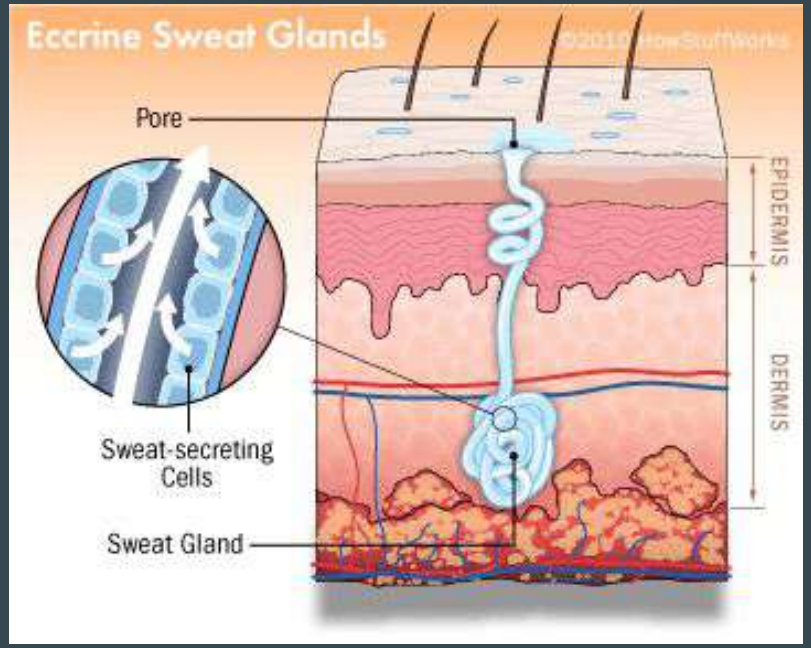
SC is measured in microsiemens (mS) with a device equipped with two electrodes to be applied on the skin



Measure of physiological signals - GSR

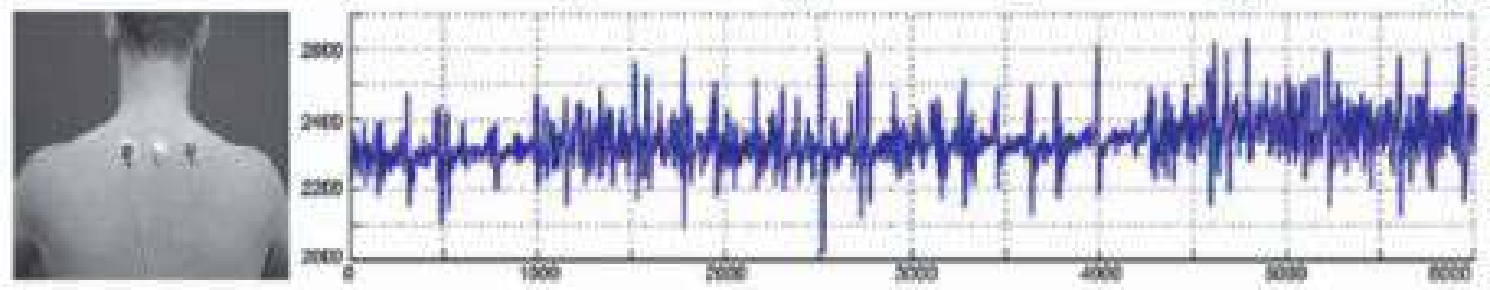
Varies with the level of skin sweating.

Sweat glands are activated by the sympathetic nervous system, therefore is a good indicator of arousal.



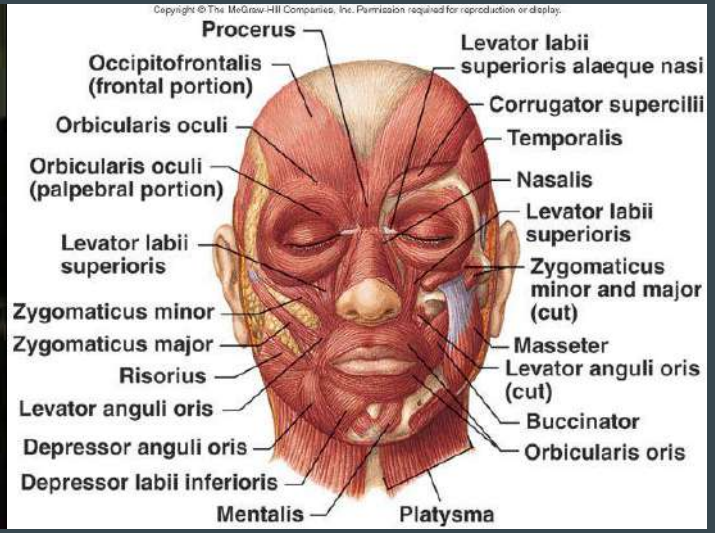
Measure of physiological signals - GSR

Electromyogram (EMG) measures muscle activity by detecting surface voltages that occur when a muscle is contracted.



Measure of physiological signals - EMG

Surface Electromyogram (sEMG) requires the application of electrodes to the skin.

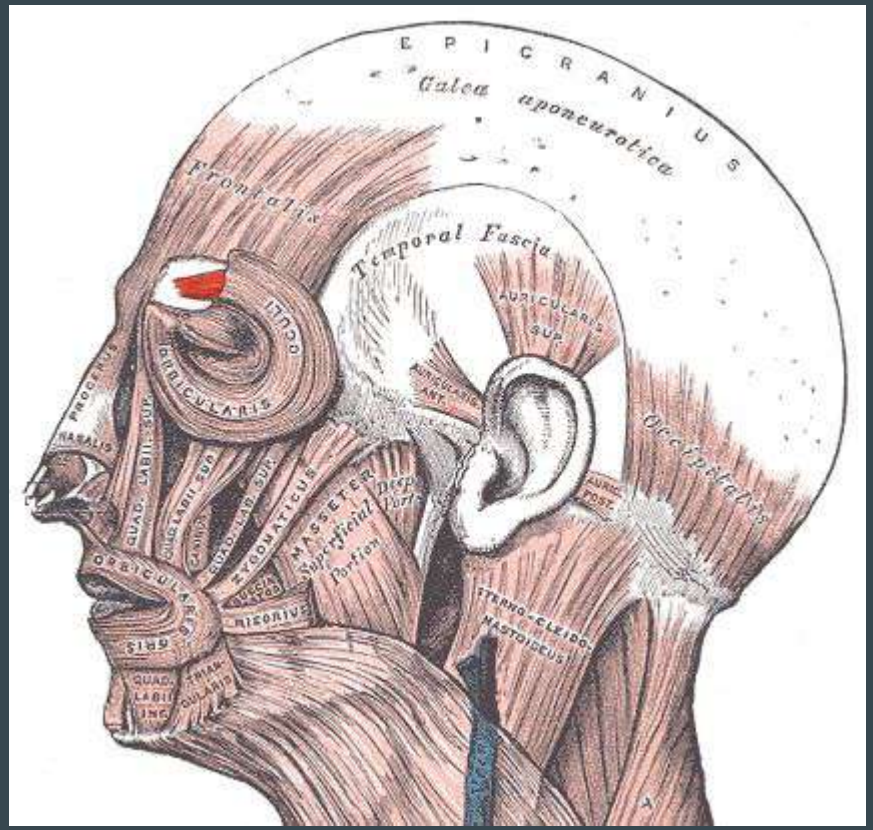


Measure of physiological signals - EMG

Corrugator supercilii muscle

Lowers the eyebrow and is involved in producing frowns.

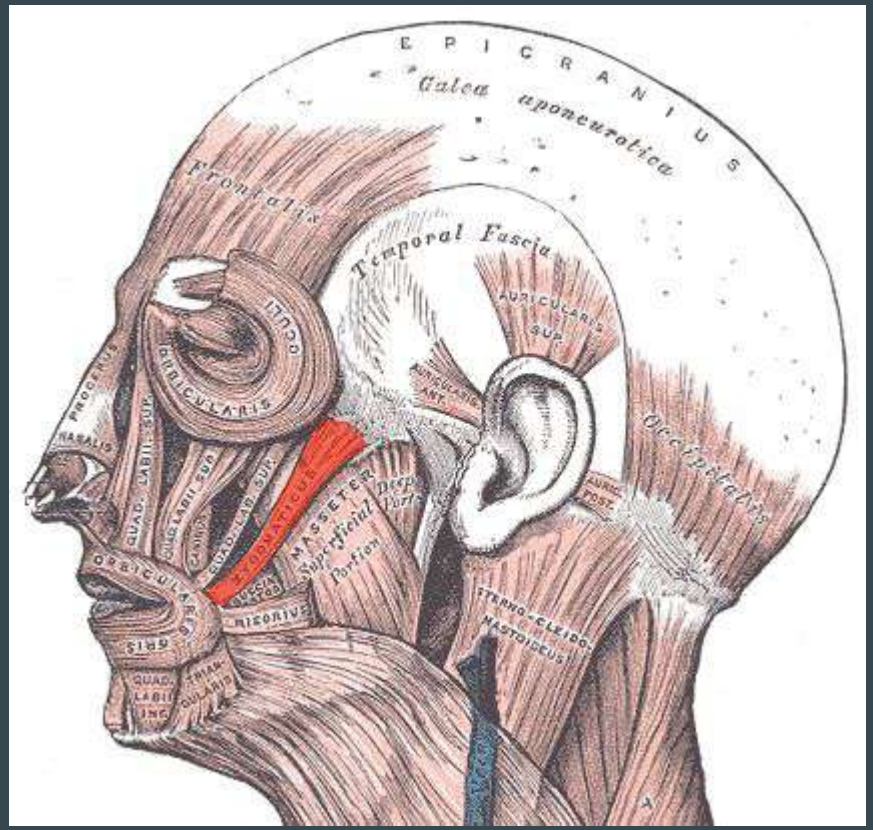
Varies inversely with the emotional valence.



Measure of physiological signals - EMG

Zygomaticus major muscle

Controls smiling and is said to be positively associated with positive emotional valence.



Measure of physiological signals - EMG